Informatique quantique 2

Grover’s Algorithm

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Plan

Part 1 :

1- Introduction to Grover's Algorithm:

- Complexity
- Main idea behind Grover's Algorithm

2- Grovers Algorithm:

- Components of the algorithm
- Geometric demonstration

3- Toy example:

- 3 qubits toy example

4- Grover's algorithm application:

- An example of the Grover's algorithm on the MIS algorithm

5- Introduction in quantum error correction :

- Simple bitflip codes
- What's behind more complex codes

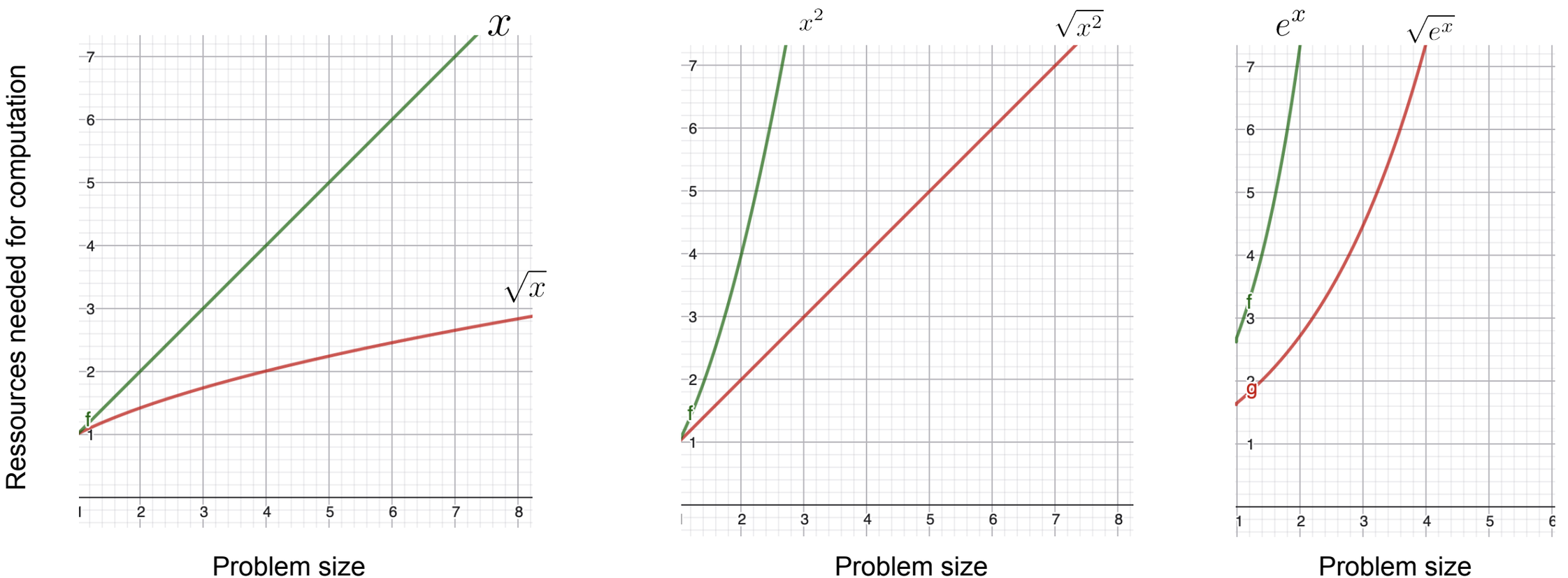
Grover’s Algorithm Introduction

Introduction

<u>Introduction on Grover's algorithm (Quadratic speedup):</u>

- To understand Grover, go to this video: https://www.youtube.com/watch?v=RQWpF2Gb-gU
- Known as the Quantum search algorithm (unstructured search)
- Aim: find and amplify a solution among a search space

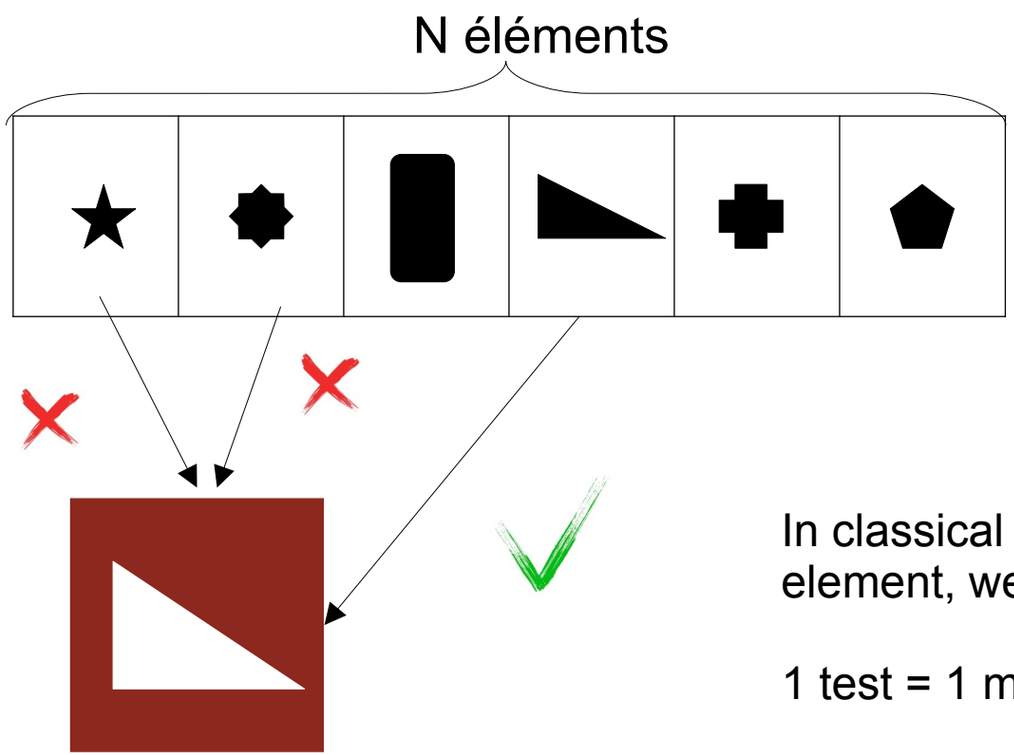
Quadratic Speedup characterization:

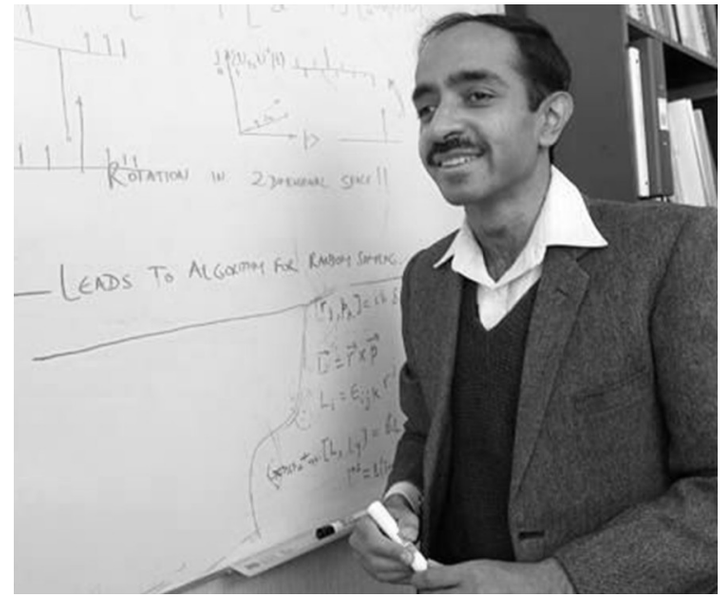


Introduction

<u>Main Idea of Grover's Algorithm:</u>

- <u>Scientific paper (1996) :</u> https://arxiv.org/abs/quant-ph/9605043





In classical computing, to be sure to find the right element, we have to do N tests in the worst case.

1 test = 1 mathematical operations

Complexity of classical algorithms:

$$O(N)$$

Introduction

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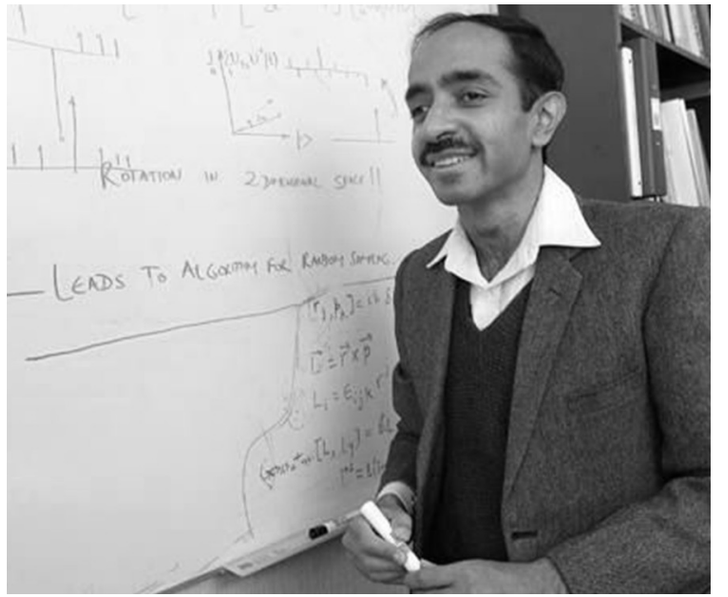
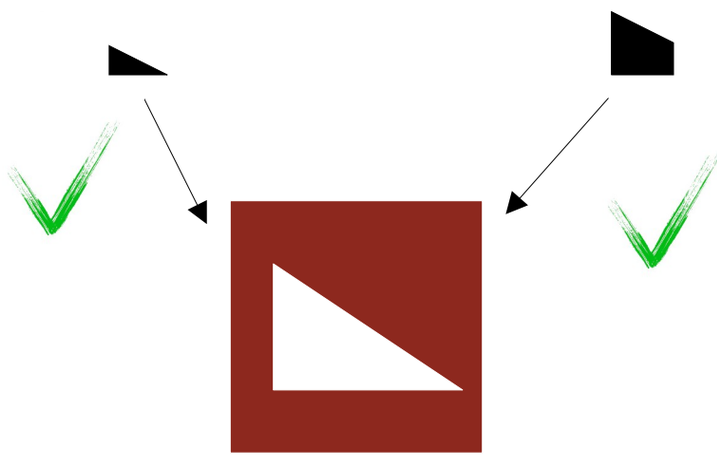


Complexity of Grover's algorithm : $O(\sqrt{N})$

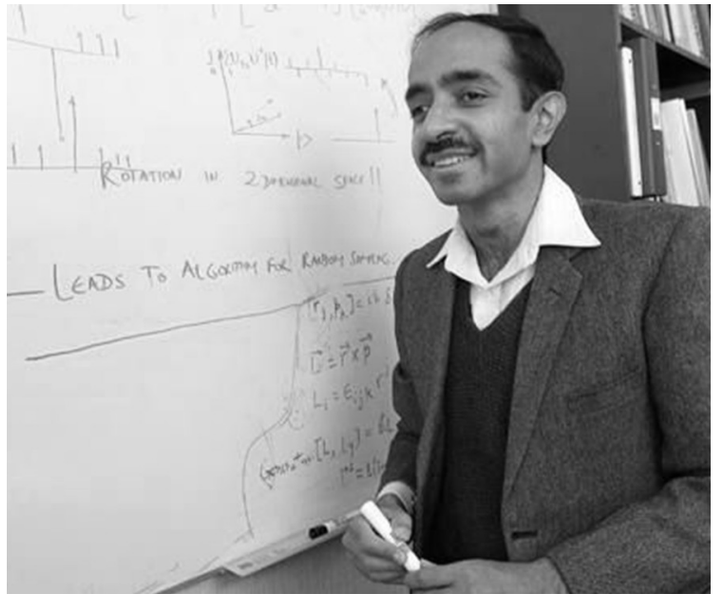
We do not know precisely the object before doing the search
We only know some rules the validated object should respect
The « database » of objects is not represented explicitely (would be too long to read the input DB and then convert it to be compatible with Grover's algorithm)

The Grover's algorithm do not process all the solutions in parallel otherwise we would be in $O(1)$

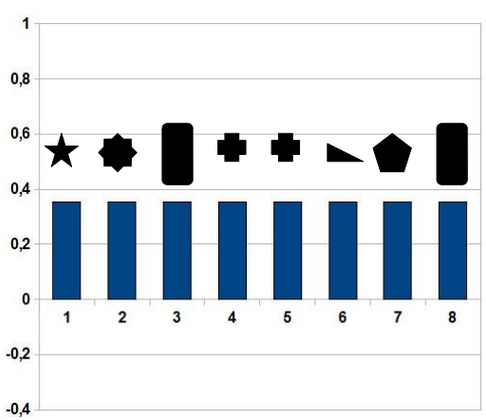


Introduction

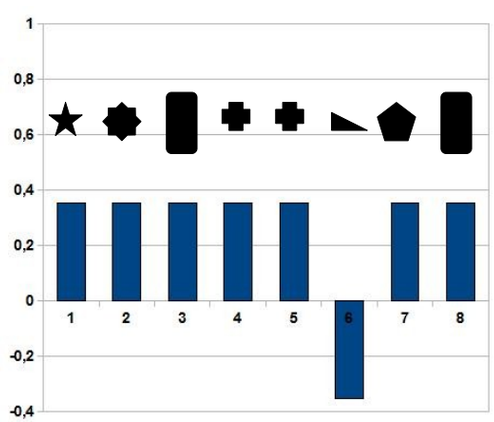
<u>Main Idea of Grover's Algorithm:</u>



- Global process of the algorithm:

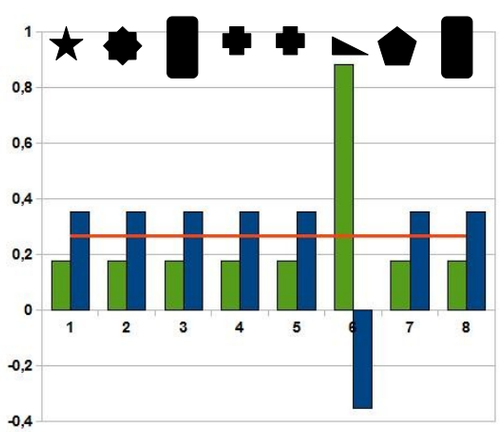


1 : Initial state

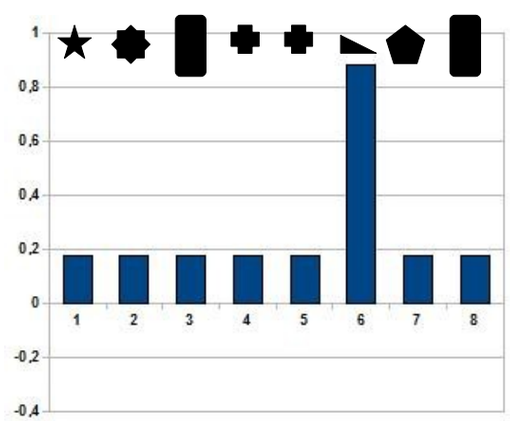


2 : Application of the oracle

=> invert the amplitude of objects validating the search rules



3 : Mirror around the average of the amplitudes of the initial state



4 : The amplitude of the state of interest is amplified

Répéter $O(\sqrt{N})$

Source : https://fr.wikipedia.org/wiki/Algorithme_de_Grover

Introduction

<u>Grover's algorithm: not as straightforward as it seems:</u>

Let's consider a table of size N, we are looking for a single element in this table. Does the Grover's algorithm find the element in time $O(\sqrt{N})$?

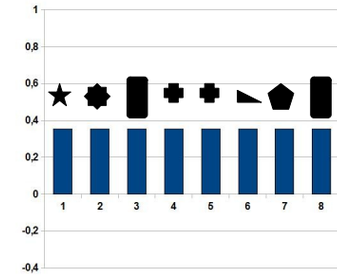
<u>The big issue:</u>
We need to compute the oracle represented by a unitary U.
=> How can we encode U by not reading all the elements of the table (equivalent to the classical search complexity $O(N)$) ?

If we find some tricks to encode the table without reading it entirely, U will have a depth that scale at least as $O(\log(N))$ or $O(N)$ which cancels any possible speedup provided by the Grover's algorithm.

Grover is **only interesting** if:
- The oracle is given as a black box (we do not consider its complexity)
- The unitary U is easily computed

Grover’s algorithm

Grover’s algorithm



1 : Initial state

<u>Initial state:</u>

- The Grover’s algorithm, for guaranteeing a quadratic speedup, assumes that the initial state is encoded on a **logarithmic number of qubits**.

- For general types of input (images, or data), this encoding step is not possible because it’s too time consuming (require linear decomposition of unitaries which generate very deep circuits).

- As a general fact, we will consider that the input state is just an **equal superposition** of the **states** of the **computational basis** with $n = \log_2(N)$:

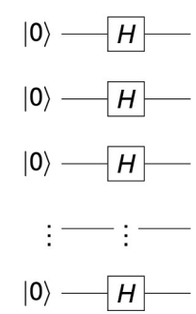
$$\frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle$$

with $|x>$ the binary encoding associated to the variable x

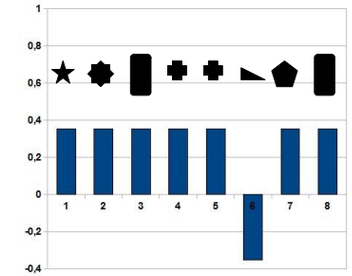
- Example on two qubits:

$$\frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle$$

This state is prepared with a wall of Hadamard gates on the n qubits



Grover’s algorithm



2 : Application of the oracle

<u>Grover’s Oracle:</u>

- The rules for selecting the right object is represented by a black box called the oracle

$$f : \{0,1\}^n \rightarrow \{0,1\}$$

If the input satisfies the right criterion: $f(x) = 1$
Othwerwise: $f(x) = 0$

<u>Assumption:</u> We assume that only one input lead to $f(x) = 1$

The function f is a subroutine which can be applied to the register of qubits as a unitary operator which acts as follows:

$$U\,|x\rangle = -\,|x\rangle$$

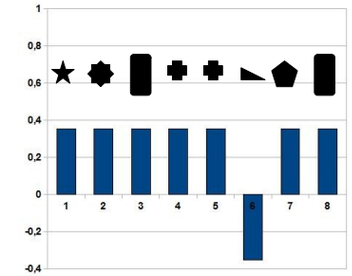
If the state x respects the criterion specified by the oracle U

$$U\,|x\rangle = |x\rangle$$

If the state x respects does not respect the criterion

=> The state of interest has its phase which is flipped by the oracle

Grover's algorithm



2 : Application of the oracle

<u>Grover's Oracle:</u>

- Considering the following binary function:

$$f : \{0,1\}^n \rightarrow \{0,1\}$$

- The application of the oracle on the quantum state can be represented as:

$$U\,|x\rangle = (-1)^{f(x)}\,|x\rangle$$

If the solution does not satisfy the search criterion

$$(-1)^0\,|x\rangle = 1\,|x\rangle = |x\rangle$$

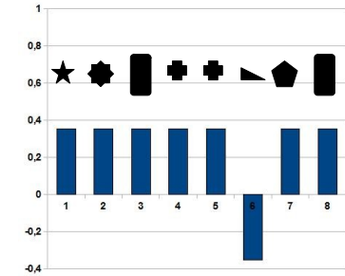
If the solution satisfies the search criterion

$$(-1)^1\,|x\rangle = -1\,|x\rangle = -\,|x\rangle$$

- The oracle is called several times during the Grover's algorithm.

- The complexity of the algorithm is measured in terms of the number of calls to the oracle. We call this type of complexity the **query complexity**.

Grover’s algorithm

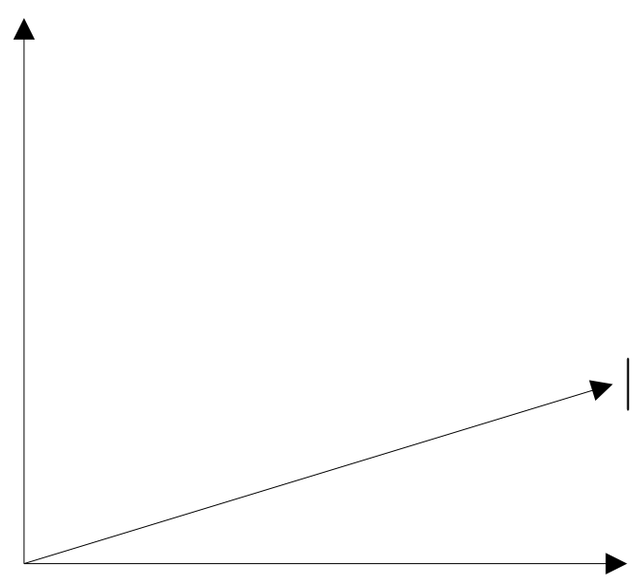


2 : Application of the oracle

<u>Grover’s Oracle:</u>

- Geometrical interpretation of the application of the Oracle:
  We only consider a space that spans the states representing the solution identified by the oracle $|sol\rangle$ and a state $|s'\rangle$ that contain all the states not in the solution, which is then orthogonal to $|sol\rangle$.

$$|sol\rangle = \frac{1}{\sqrt{N}} \sum_{x|f(x)=1} |x\rangle$$

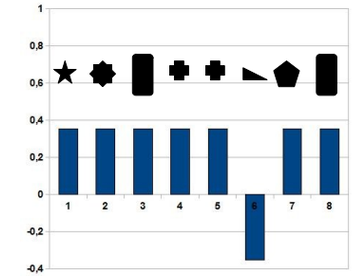


The equal superposition |s> can then be represented in the space spanned by |sol> and |s’>

$$|s\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle$$

$$|s'\rangle = \frac{1}{\sqrt{N-1}} \sum_{x \neq w} |x\rangle$$

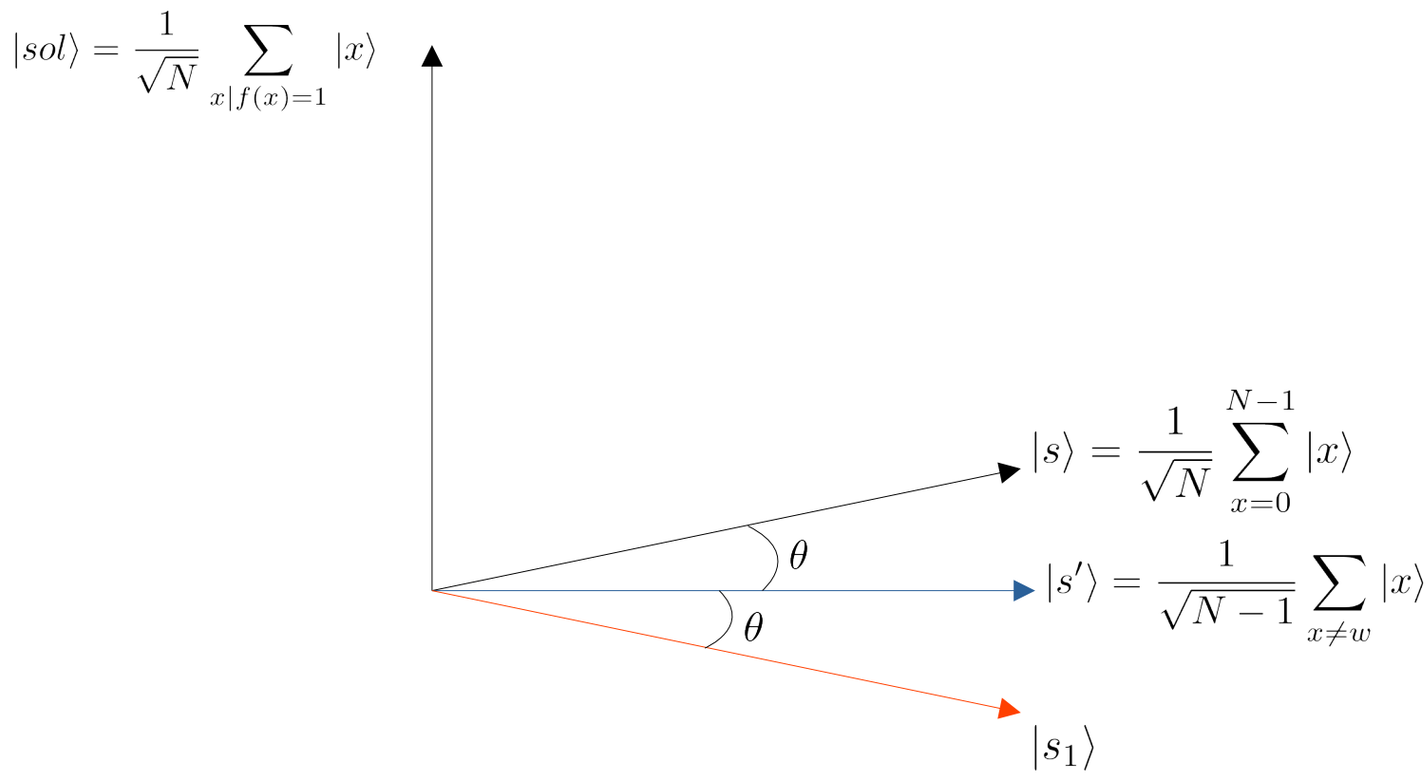
Grover's algorithm



Application of the oracle

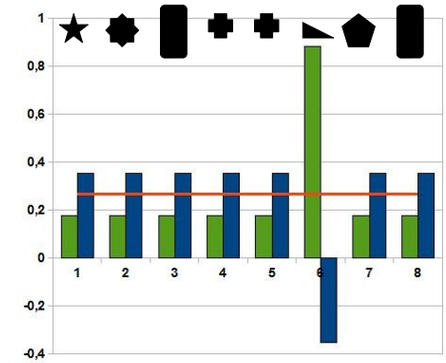
Grover's Oracle:

$$|s_1\rangle = U_{\text{oracle}} |s\rangle$$



U_{oracle} is a reflection of the $|s'\rangle$ axis.

Grover's algorithm



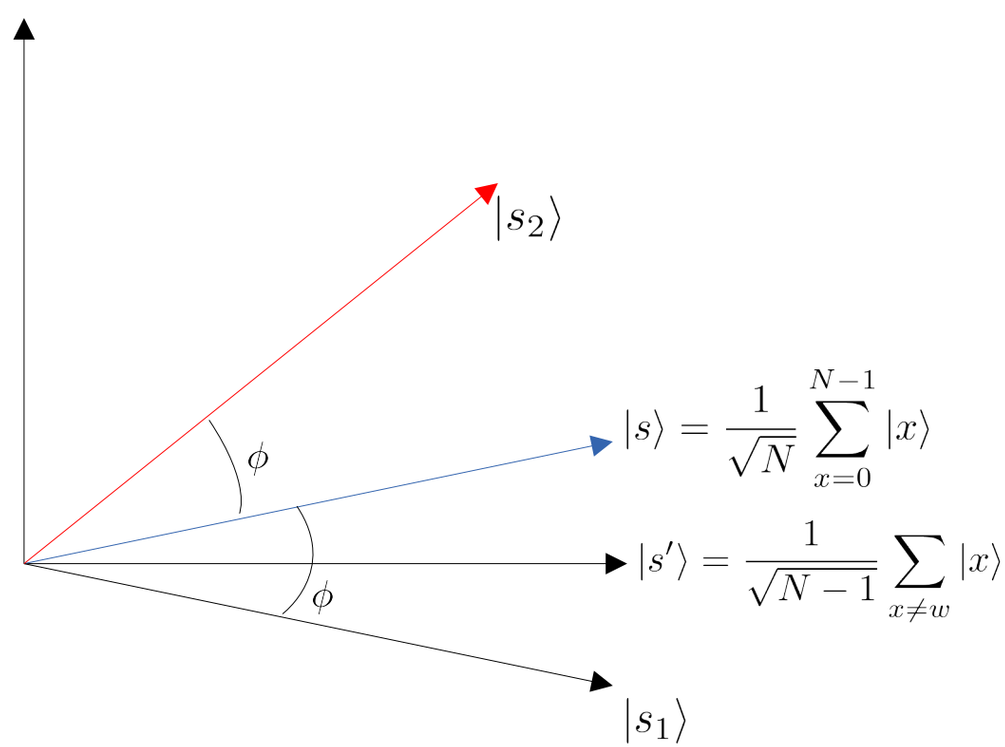
Mirror around the average of the amplitudes of the initial state

$U_{amplify}$ does the reflection of $|s\rangle$ axis

<u>Grover's Oracle:</u>

$$|s_2\rangle = U_{\text{amplify}} |s_1\rangle$$

$$|sol\rangle = \frac{1}{\sqrt{N}} \sum_{x|f(x)=1} |x\rangle$$



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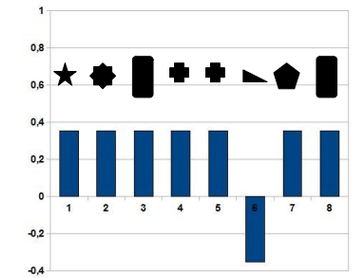


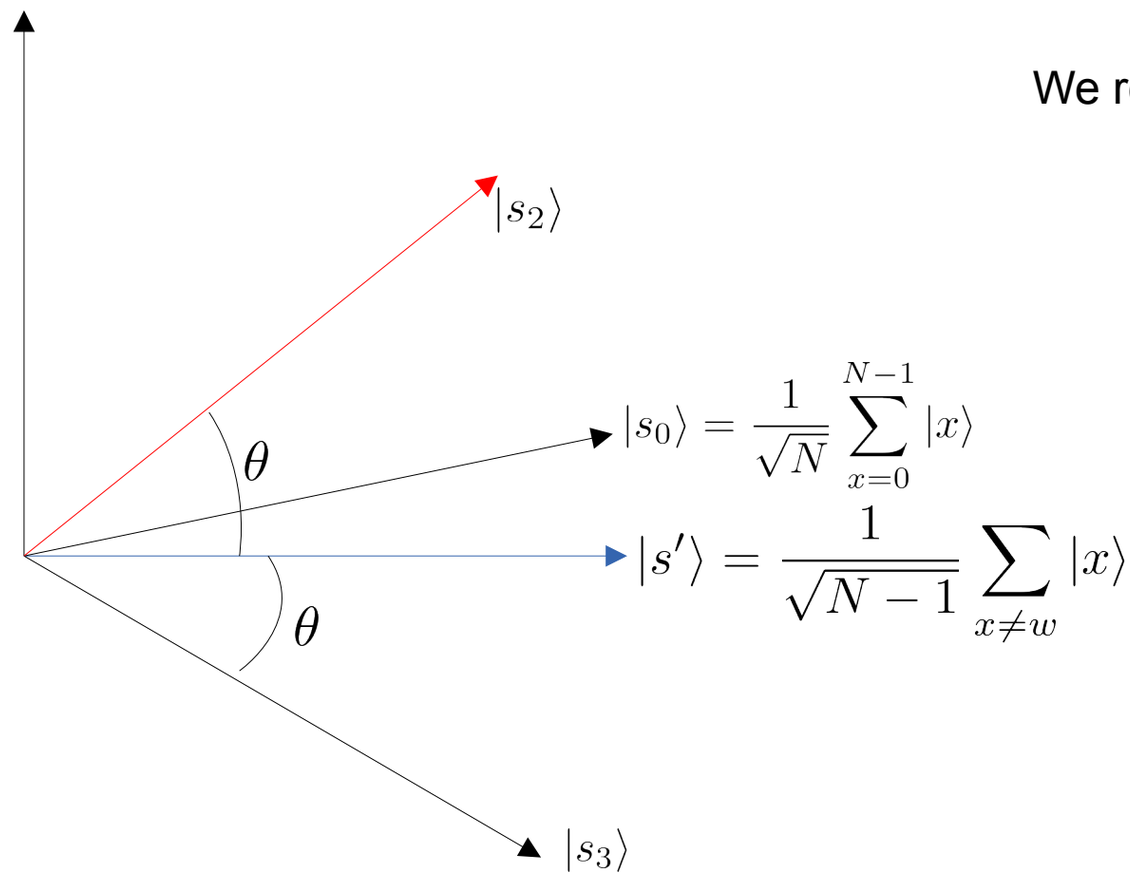
Application of the oracle

<u>Grover's Oracle:</u>

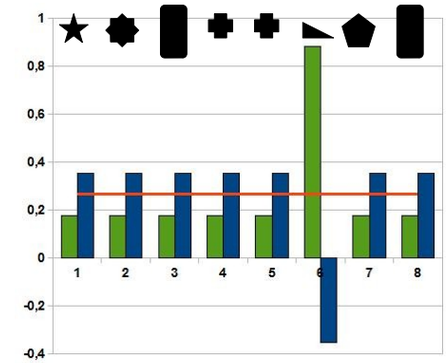
$$|s_3\rangle = U_{\text{oracle}} |s_2\rangle$$

$$|sol\rangle = \frac{1}{\sqrt{N}} \sum_{x|f(x)=1} |x\rangle$$

We repeat the same steps...



Grover's algorithm

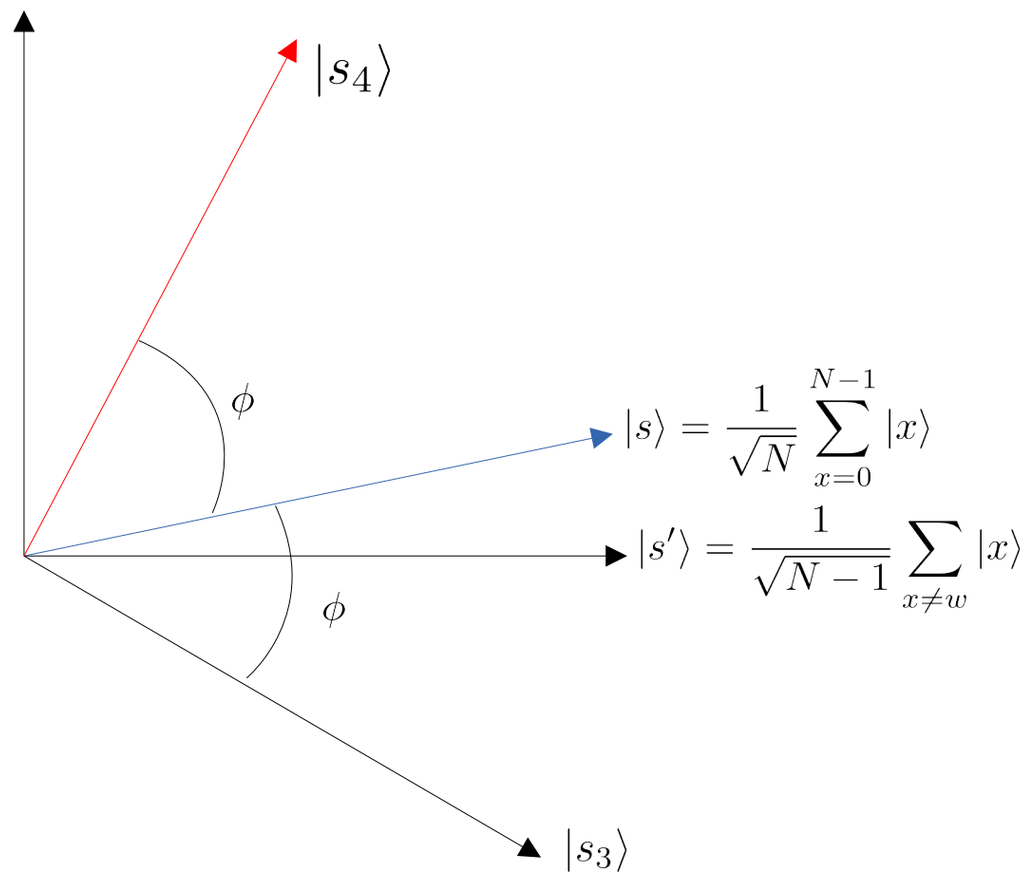


Mirror around the average of the amplitudes of the initial state

Grover's Oracle:

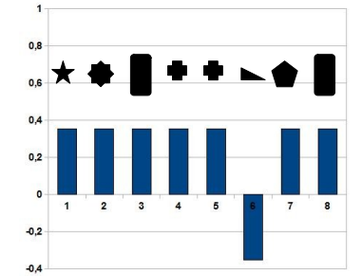
$$|s_4\rangle = U_{\text{amplify}} |s_3\rangle$$

$$|sol\rangle = \frac{1}{\sqrt{N}} \sum_{x|f(x)=1} |x\rangle$$



We repeat the same steps...

Grover's algorithm

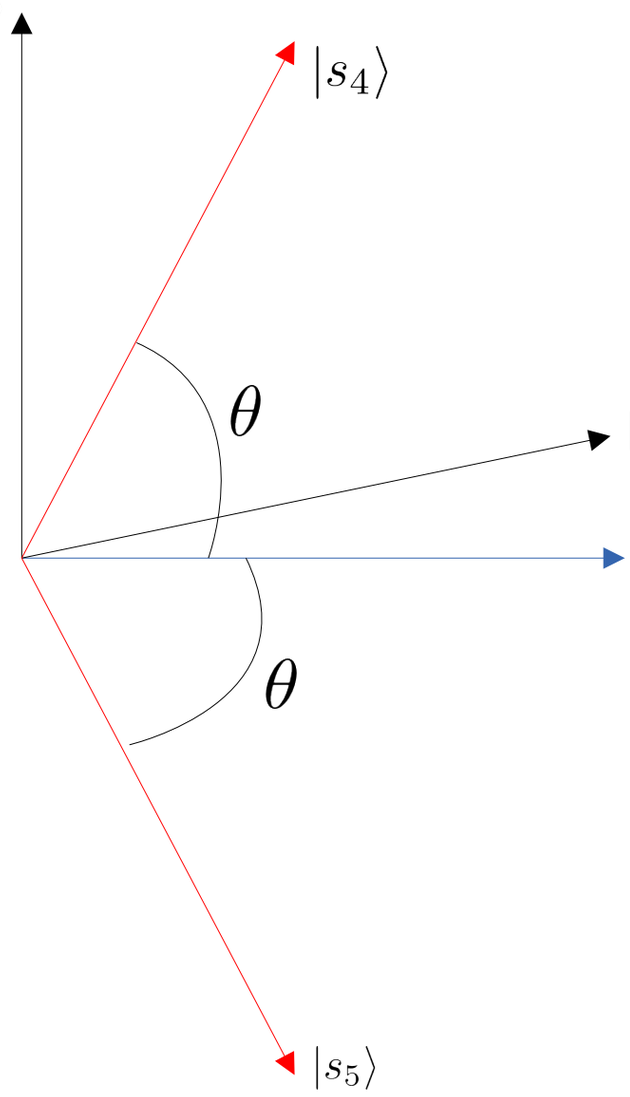


Application of the oracle

Grover's Oracle:

$$|sol\rangle = \frac{1}{\sqrt{N}} \sum_{x|f(x)=1} |x\rangle$$

$$|s_5\rangle = U_{\text{oracle}} |s_4\rangle$$



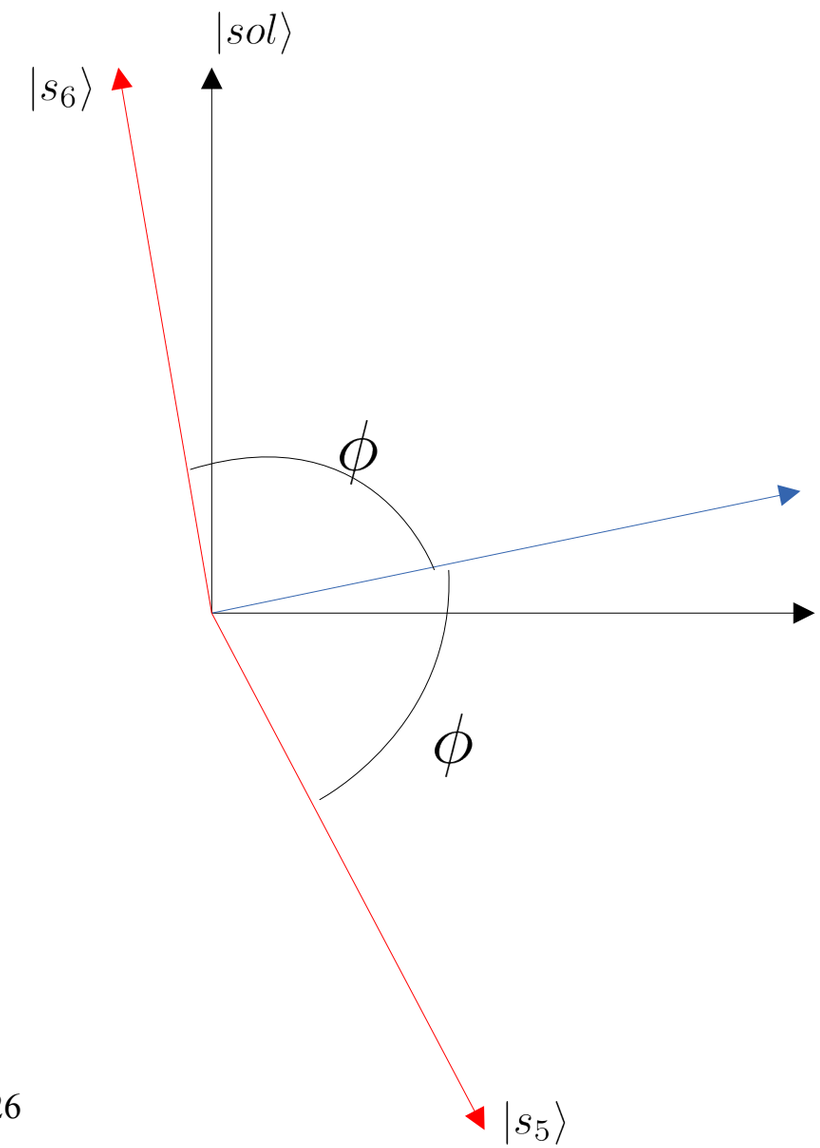
$$|s\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle$$

$$|s'\rangle = \frac{1}{\sqrt{N-1}} \sum_{x \neq w} |x\rangle$$

We repeat the same steps...

Grover's algorithm

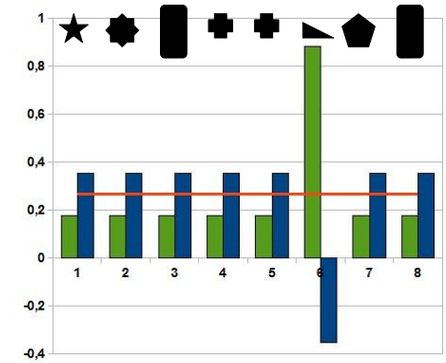
<u>Grover's Oracle:</u>



$$|s_6\rangle = U_{\text{amplify}}\,|s_5\rangle$$

$$|s\rangle = \frac{1}{\sqrt{N}}\sum_{x=0}^{N-1}|x\rangle$$

$$|s'\rangle = \frac{1}{\sqrt{N-1}}\sum_{x\neq w}|x\rangle$$



Mirror around the average of the amplitudes of the initial state

We repeat the same steps...

If we continue the reflection process, we will decrease the probability of observing the solution $|sol\rangle$

The complexity $O(\sqrt{N})$ comes from the number of reflections required to converge to the solution

Grover's algorithm

<u>Number of iterations:</u>

Important theorem:
Given a unitary matrix U (specified by a quantum circuit), there an integer n for which:

$$U^n \approx I$$

=> It means that if we do too much iterations, amplitudes of the amplified state will lower at some point.

=> The ideal number of Grover's iteration must be precisely pre-computed before running the quantum algorithm to stop at the right time, which is when the amplitude of the state satisfying the criterion is maximum.

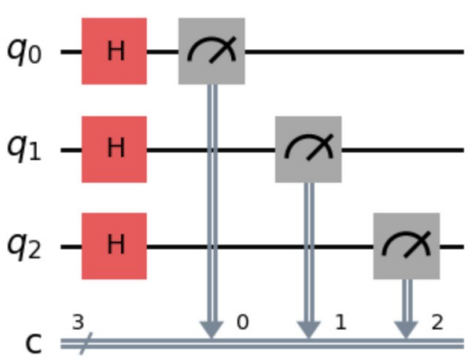
The number of rotations to maximize the probability of observing $|sol\rangle$ scales as $\frac{4\pi}{3}O(\sqrt{N})$

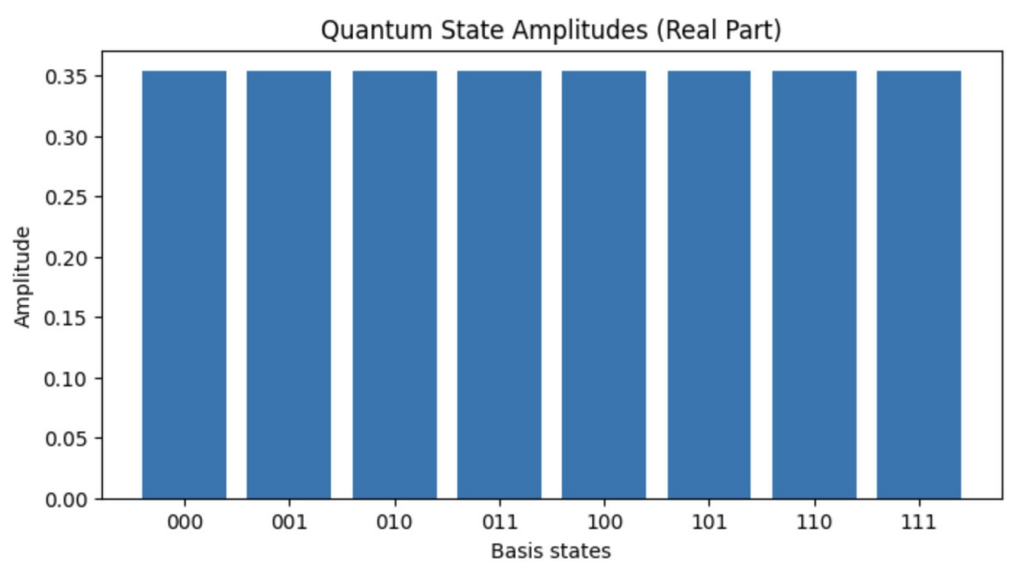
Grover’s algorithm: a toy example

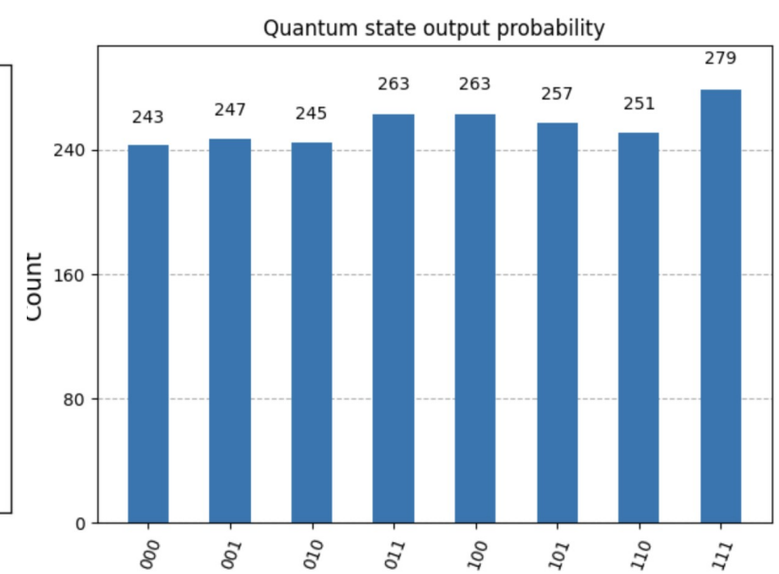
Toy example

<u>Two qubits toy example:</u>

The initial state is an equal superposition of the qubits in the computational basis:







$$|\psi\rangle = \frac{1}{\sqrt{8}}|000\rangle + \frac{1}{\sqrt{8}}|001\rangle + ... + \frac{1}{\sqrt{8}}|111\rangle$$

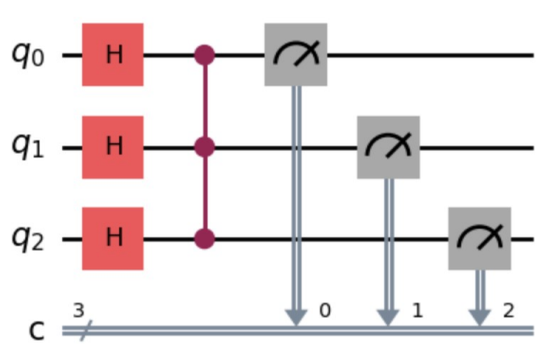
Toy example

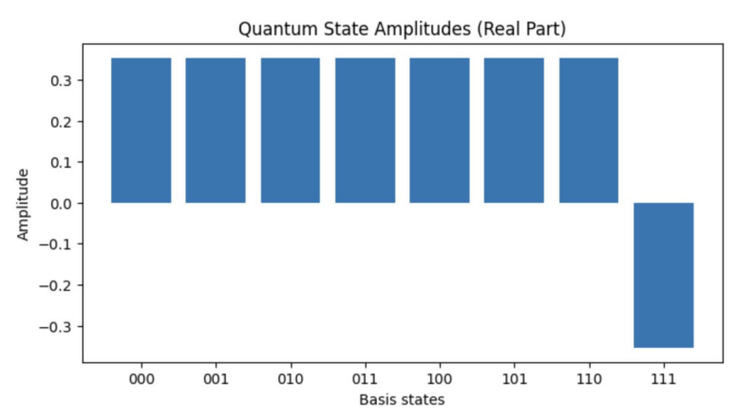
<u>Two qubits toy example</u>:
Let's say the state of interest marked by the oracle is $|111\rangle$:

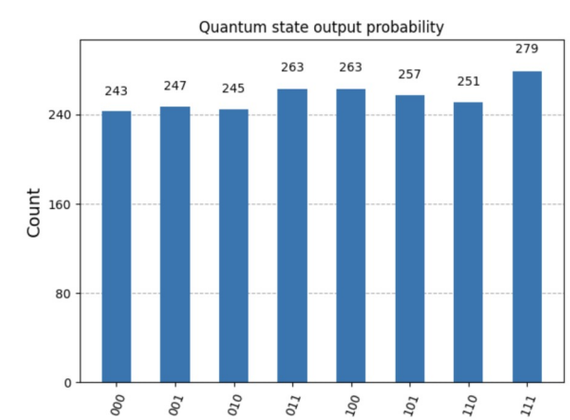
Which quantum operation could be used to flip the phase of this state ?

$$CCZ\ q_0, q_1, q_2 = I \otimes I \otimes |0\rangle\langle 0| + CZ \otimes |1\rangle\langle 1| = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \end{pmatrix}$$

The quantum circuit now looks like (the amplitude flip do not change the sampling output):







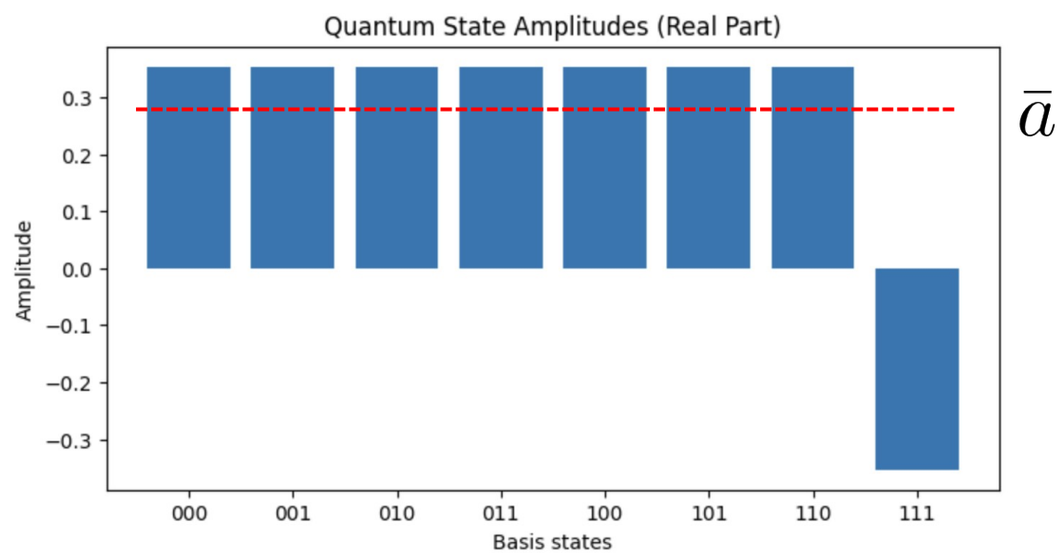
$$|\psi\rangle = \frac{1}{\sqrt{8}}|000\rangle + \frac{1}{\sqrt{8}}|001\rangle + \ldots - \frac{1}{\sqrt{8}}|111\rangle$$

Toy example

<u>Two qubits toy example</u>:
Diffusion operator: The diffusion operator does an inversion about the mean of the amplitudes:

$$\bar{a} = \left(\left(\frac{1}{\sqrt{8}} \right) \times (7-1) \right) /8 \approx 0.27$$

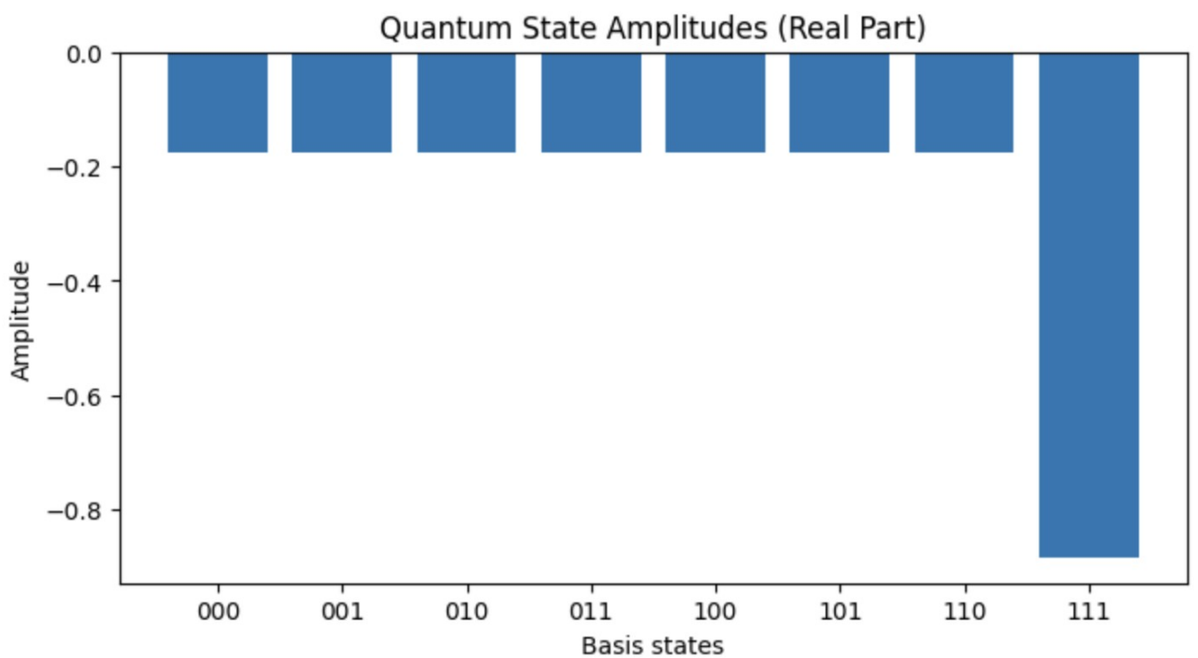
Mean before the diffusion operator:

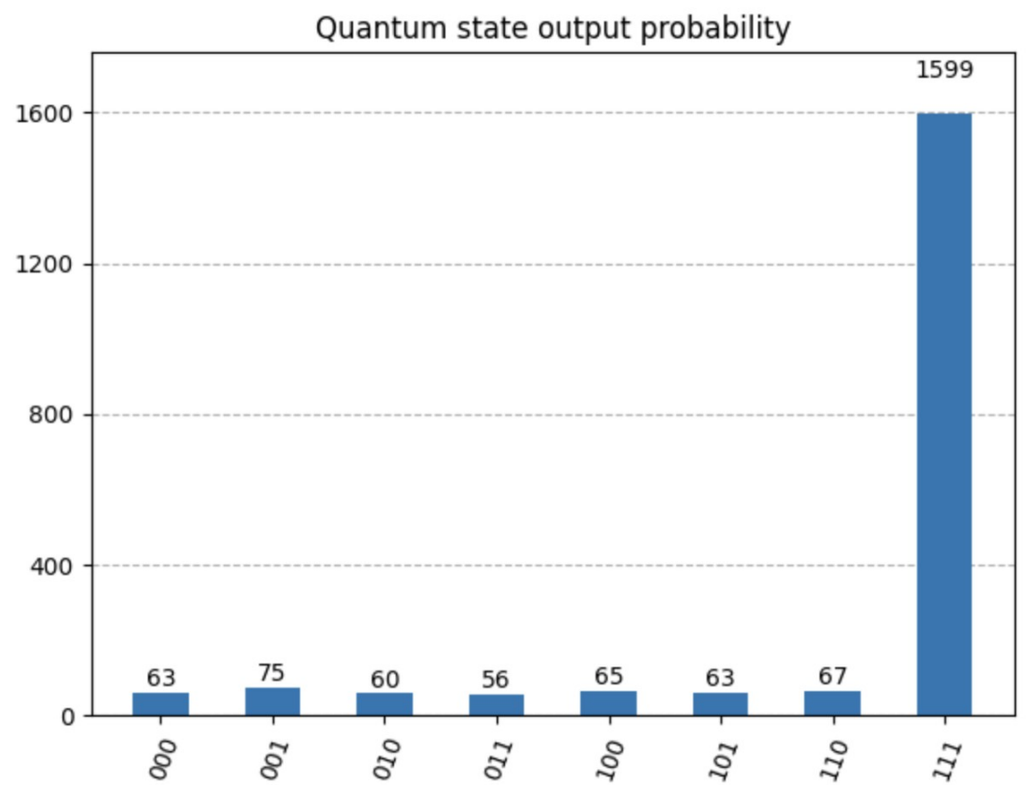


After the diffusion operator, each amplitude a_i are rescaled as: $a_i' = 2\bar{a} - a_i$

Toy example

<u>Two qubits toy example</u>:
New amplitudes and output state probability:





Amplitude after diffusion operator of the state of interest: $0.27 \times 2 + 0.35 \approx 0.89$

*at a global phase

Amplitude after diffusion operator of the other states: $0.27 \times 2 - 0.35 \approx 0.19$

Explanation on why the global phase does not matter:
https://quantumcomputing.stackexchange.com/questions/40713/step-by-step-explanation-of-grover-diffusion-operator-quantum-circuit-for-2-qubi

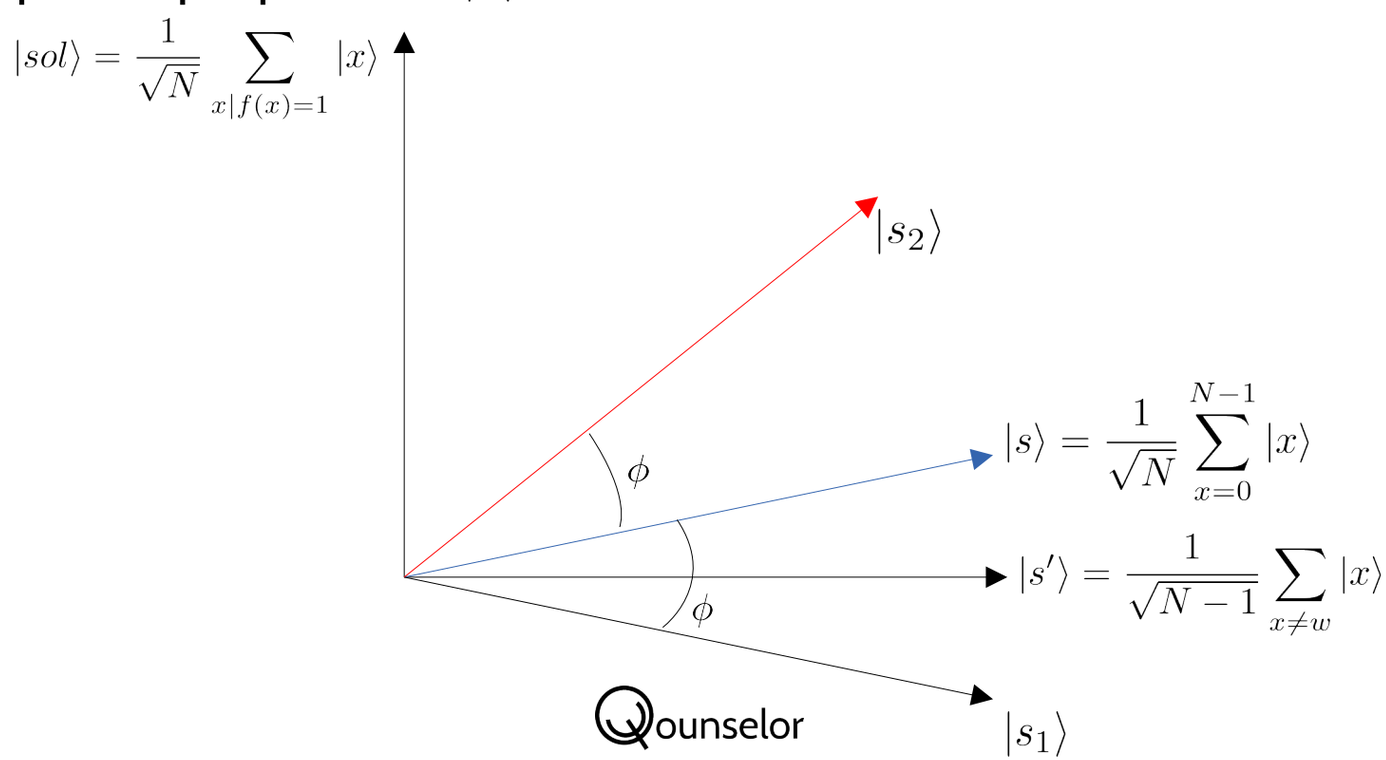
Toy example

<u>Two qubits toy example</u>:

We now apply the amplification step (this circuit is always the same):

$$H^{\otimes n}(2\left|0\right\rangle\left\langle 0\right| - I)H^{\otimes n}$$

Remember, we want to switch the amplitudes around the axis defined by the state that is the initial equal superposition $\left|s\right\rangle$:



Toy example

<u>Two qubits toy example</u>:

We now apply the amplification step (this circuit is always the same):

$$H^{\otimes n}(2\left|0\right\rangle\left\langle 0\right| - I)H^{\otimes n}$$

To do this reflection, the wall of Hadamard changes the basis of the quantum state to the Hadamard basis, such that the state $\left|s\right\rangle$ is now represented as the state $\left|000\right\rangle$

Compute $2\left|00\right\rangle\left\langle 00\right| - I_1 I_2$
What is its action considering that your state is in the Hadamard basis ?

$$2 \times \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

In general we will have:

$$diag(1, -1, -1, ..., -1)$$

The amplitudes are reflected (multiplication by the coef -1) in the Hadamard basis

Toy example

<u>Two qubits toy example</u>:

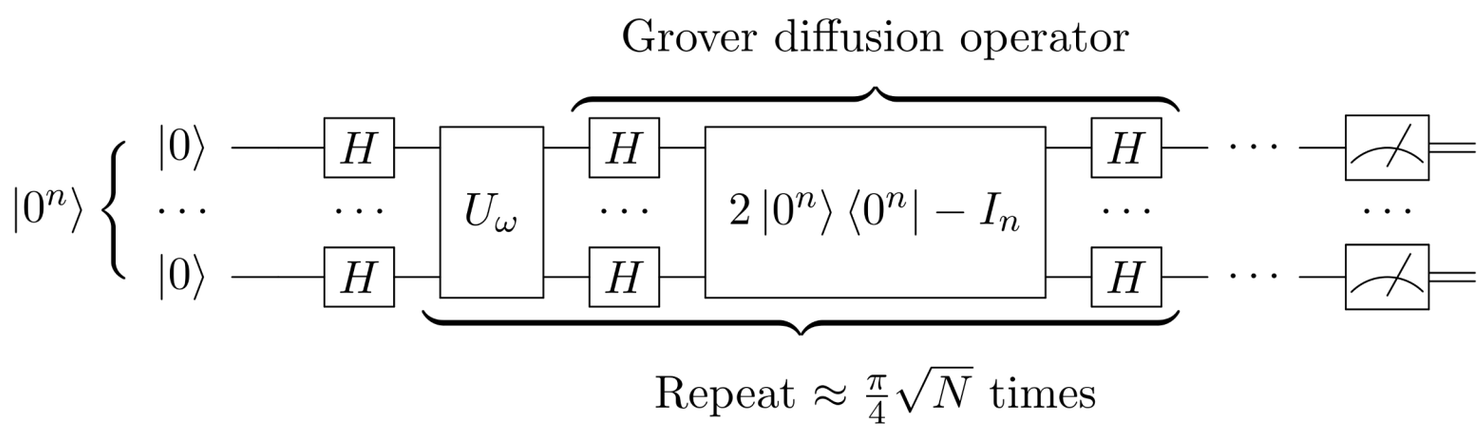
We now apply the amplification step is decomposed as:

$$H^{\otimes n}\underbrace{(2\left|0\right\rangle\left\langle 0\right| - I)}H^{\otimes n}$$

* in practice, we compute a matrix encoding $-(2\left|0\right\rangle\left\langle 0\right| - I)$ because it's easier

$$X^{\otimes n} Multi - control - ZX^{\otimes n}$$

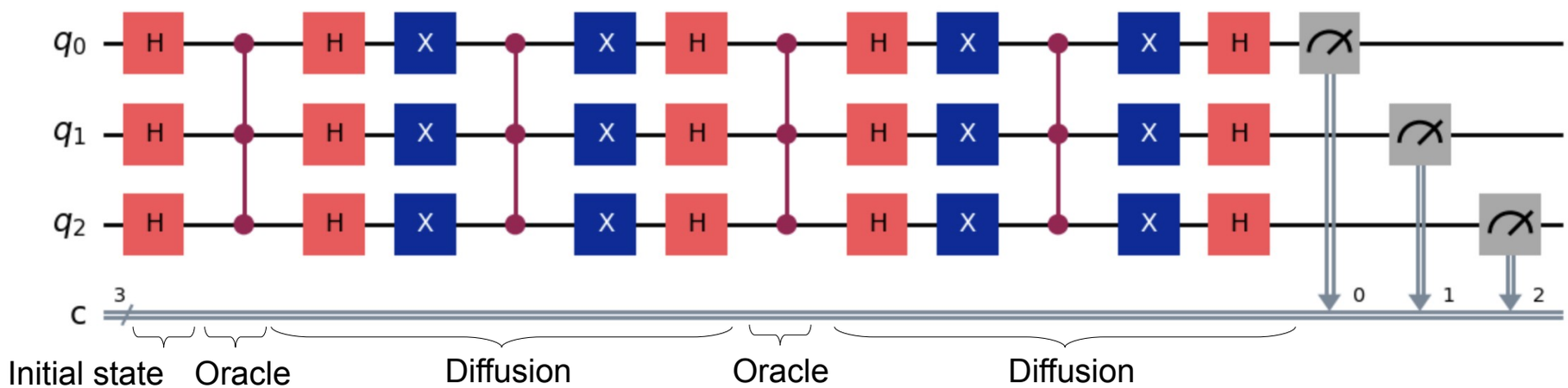
Grover's quantum circuit:



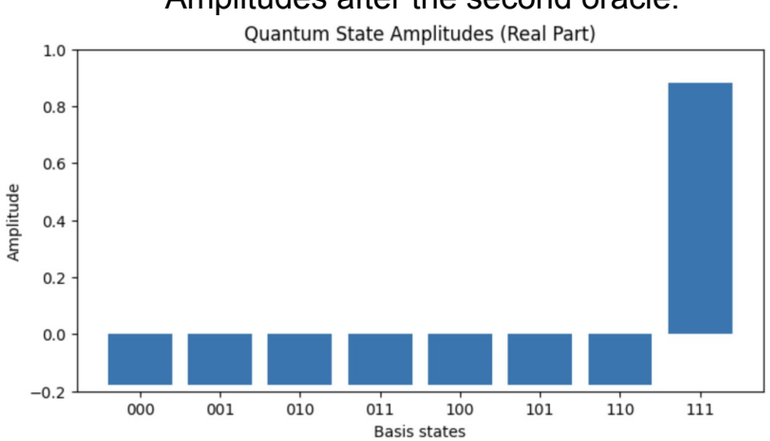
Toy example

<u>Two qubits toy example</u>:

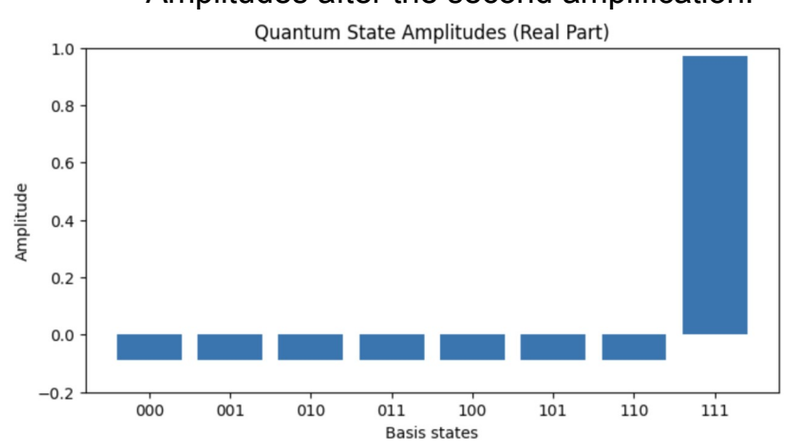
We repeat the oracle and amplification step:

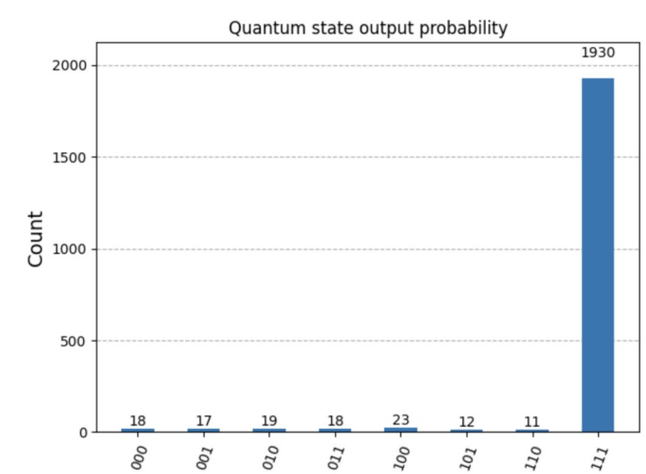


Amplitudes after the second oracle:



Amplitudes after the second amplification:

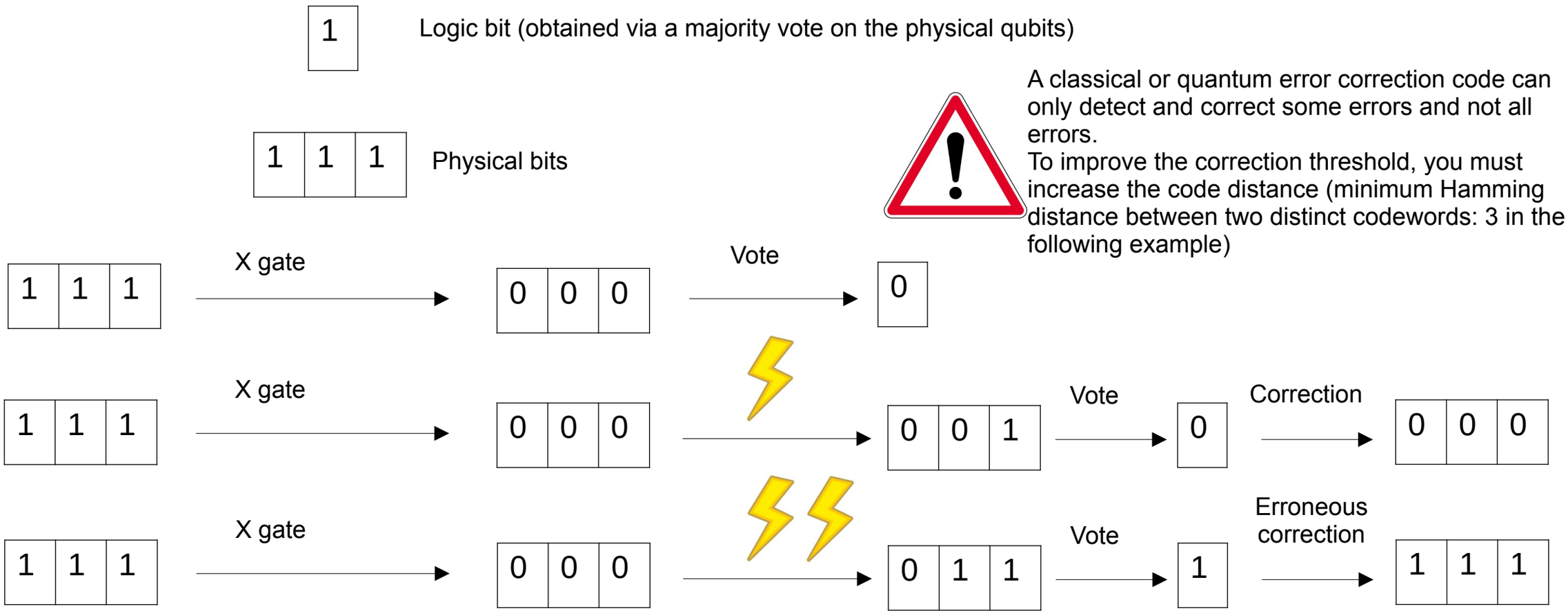




Quantum Error Correction

Quantum Error Correction

<u>Introduction to classical error correction (only compatible with digital quantum computers):</u>

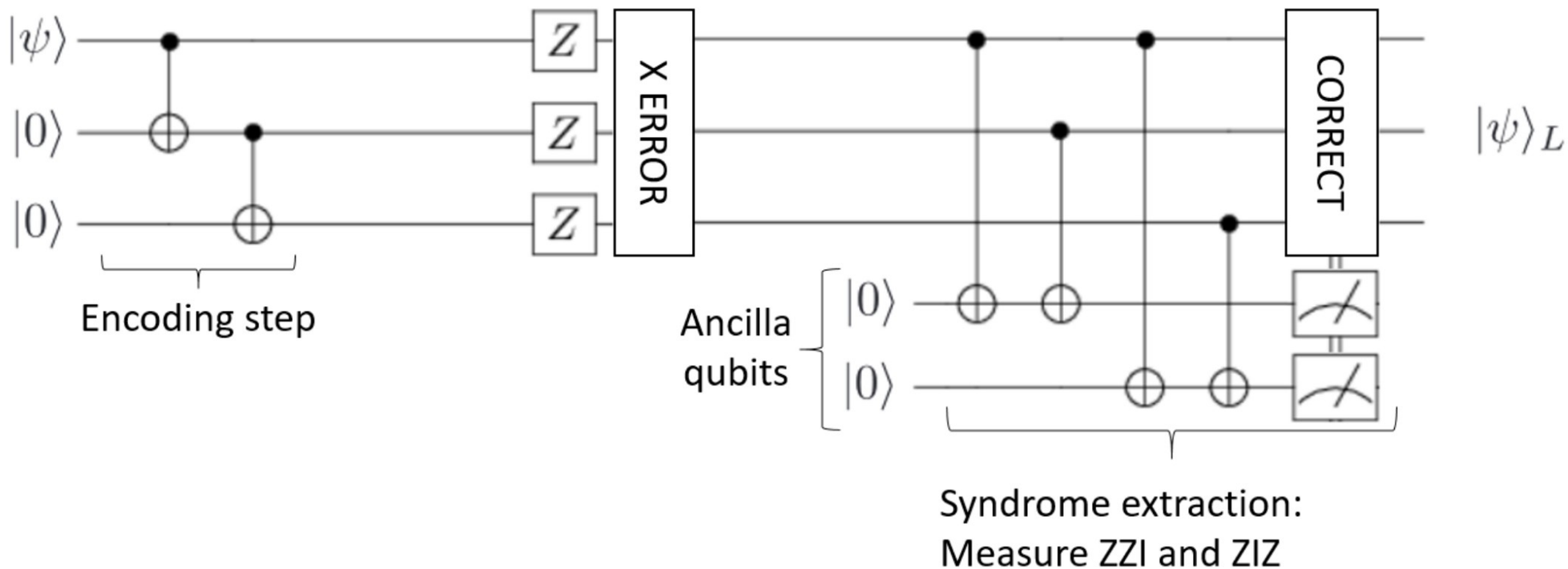


A classical or quantum error correction code can only detect and correct some errors and not all errors.
To improve the correction threshold, you must increase the code distance (minimum Hamming distance between two distinct codewords: 3 in the following example)

Quantum Error Correction

Introduction to quantum error correction:

3-qubit code (able to detect a bit flip by comparing the parity of qubits two by two):



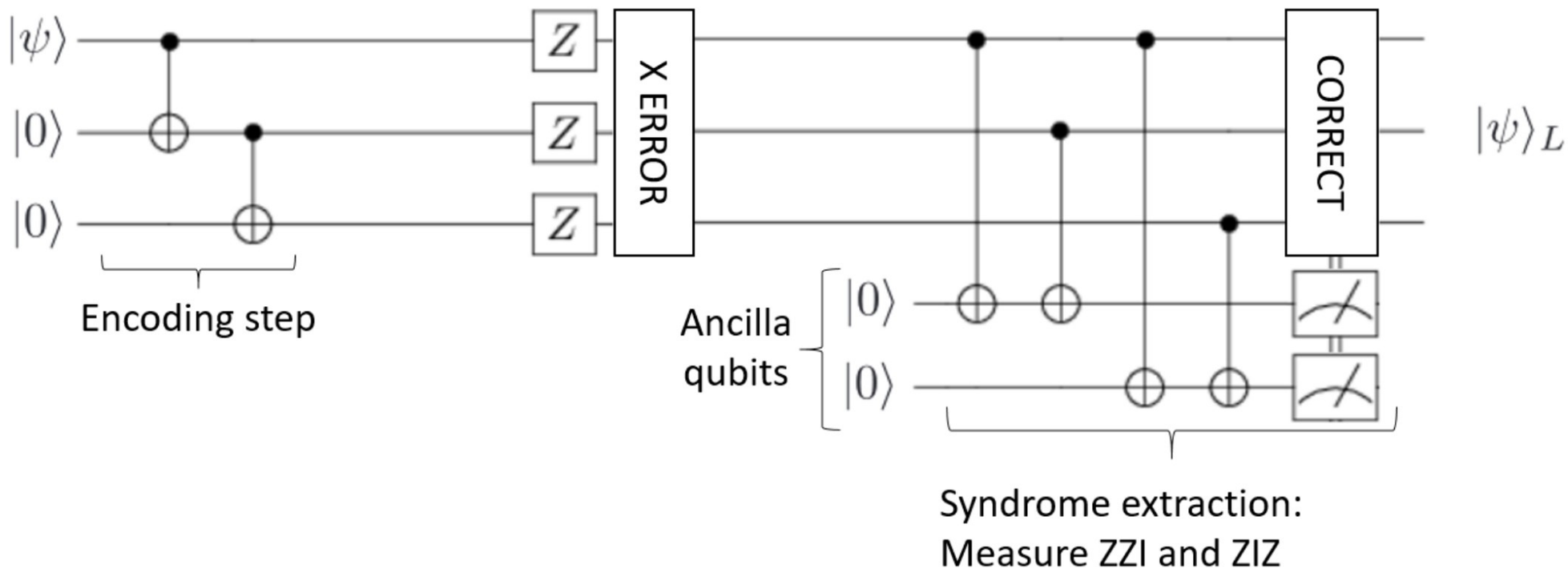
- The logical qubit $|\psi_L\rangle$ is encoded on three physical qubits.
- The random bit flip error flips the state of one qubit, which is detected by the ancilla qubits

How many bit flip errors can this code detect ?

Quantum Error Correction

Introduction to quantum error correction:

- What if the X error is not a full bit flip but just some random rotation X between $[0, \pi]$?



- The measurement of the ancilla **stabilizes** the quantum circuit meaning that:
  - Either the error appears and is detected by the syndrome extraction.
  - Either the error is cancelled and no error is detected when extracting the syndrome.

Quantum Error Correction

Introduction to quantum error correction:

- In practise, we should be able to correct errors both on x and z axis of the Bloch sphere. The bit flip quantum error correction code is then not sufficient to correct all the possible errors.

- Main idea behind quantum error correction:
  - Define a set of stabilizers (ways of indirect measurement of the qubit parity). For example for the bit flip detection scheme:

$$S = \{ZZI, ZIZ\}$$

- You stabilizers should satisfy:

$$|\psi_L\rangle = S_1 |\psi_L\rangle \qquad\qquad |\psi_L\rangle = S_2 |\psi_L\rangle$$

Quantum Error Correction

<u>Introduction to quantum error correction:</u>

- Imagine now you that you have an error modeled by some operator E.

If the error commute with your stabilizer:

$$[S_i, E] = S_i E - E S_i = 0$$

Then the error remains undetected (+1 eigenstate of S):

$$S(E\,|\psi_L\rangle) = ES\,|\psi_L\rangle = E\,|\psi_L\rangle$$

On the other hand, if the error anti-commute with the stabilizer (-1 eigenstate of S):

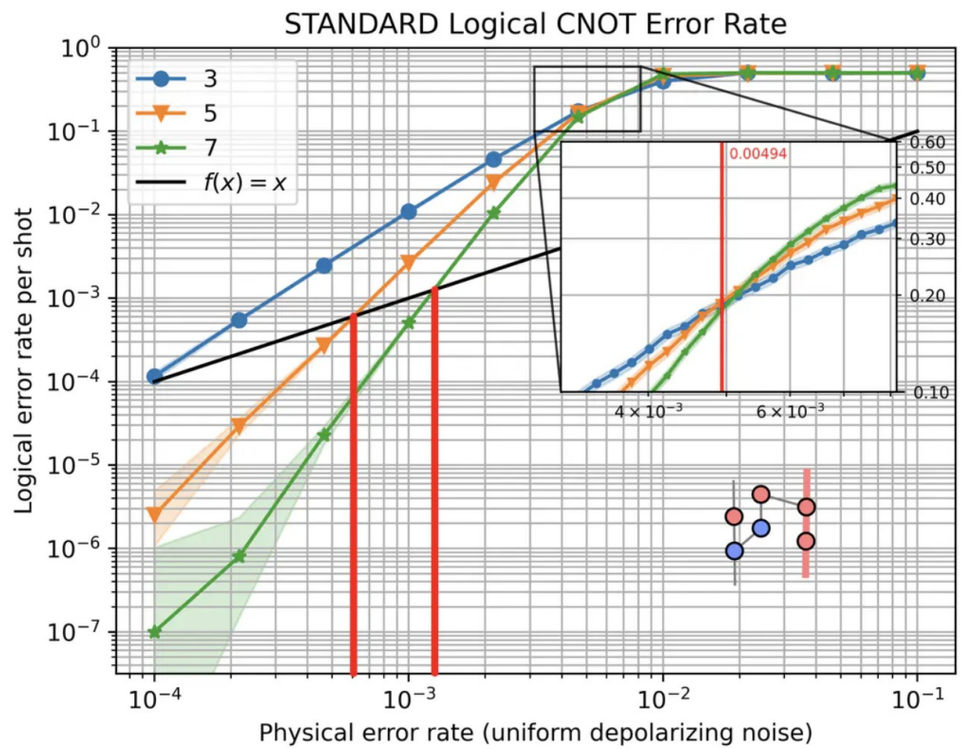
$$S(E\,|\psi_L\rangle) = ES\,|\psi_L\rangle = -E\,|\psi_L\rangle$$

=> The error is detected with the stabilizer.
<u>Aim of QEC:</u> Build sets of intelligent stabilizers that maximize the number of errors being detected.

Quantum Error Correction

<u>Introduction to quantum error correction:</u>

Code threshold with respect to the distance of the code (proportional to the number of physical qubits used to encode the logical qubit)



Source: https://adrien.suau.me/post/2025-03-25-tqec-introduction/

- Documentation if you want to explore the subject more deeply:
  - D. Gottesman (one of the father of QEC): https://arxiv.org/pdf/quant-ph/0004072
  - Surface code – A. Fowler: https://arxiv.org/abs/1208.0928

Thank you for your attention !